Siddhartha Prasad

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I am a researcher focused on helping people write programs that behave as they intend. My research interests are informed by my time as an engineer: I have written code that doesn't do what I want it to, and I want to spare everyone else the indignity.

Education

Brown University Providence, RI *Jan 2022 – present*

Ph.D. in Computer Science

Tufts University Medford, MA *May 2016*

B.S. in Computer Science and Mathematics
Summa cum Laude; High Thesis Honors

Employment

Apple Seattle, WA

Research Intern · Human-Centered Machine Learning *May – Sept 2023*

Explored how AI and formal-methods techniques can support early childhood education.

Microsoft Redmond, WA

Software Engineer II · Applied AI *April 2018 – Sept 2021*

- Developer on the Azure Cognitive Services team, which lets developers embed AI into apps, bots, and websites.
- Built a containerized framework that allowed AI models to be run portably across operating systems and architectures.

Software Engineer · Developer Ecosystem Platform *Aug 2016 – March 2018*

- Developed APIs and features for XAML, a UI language used to build Universal Windows apps.
- Designed XAML APIs for input modalities (keyboard, gamepad, ink), accessibility, and cross-platform language support.

Software Engineering Intern · Developer Ecosystem *May – Aug 2015*

INRIA Saclay, France

Research Intern · Parsifal *June – July 2014*

Research on correctness certificates for first-order term rewriting.

Publications

Diagramming Program Values by Spatial Refinement, *PLDI 2026*
Siddhartha Prasad, Michael Tu, Karan Kashyap, Tim Nelson, Shriram Krishnamurthi

Meaningful Human-in-the-Loop Checking of GenAI Synthesis for Restricted Languages, *ECOOP 2026*
Siddhartha Prasad, Skyler Austen, Kathi Fisler, Shriram Krishnamurthi

hS: Speculative Script Reordering at Subprocess Granularity, *OSDI 2026*
Georgios Liargkovas, Di Jin, Tianyu (Ezri) Zhu, Dan Liu, A. Bolun Thompson, Anirudh Narsipur, Seong-Heon Jung, *Siddhartha Prasad*, Diomidis Spinellis, Michael Greenberg, Konstantinos Kallas, Nikos Vasilakis

A Misconception-Driven Adaptive Tutor for Linear Temporal Logic, *CAV 2025*
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
[Distinguished Paper]

Lightweight Diagramming for Lightweight Formal Methods: A Grounded Language Design, *ECOOP 2025*
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
[Distinguished Paper; Distinguished Artifact]

Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic, *FM 2024*
Ben Greenman, *Siddhartha Prasad*, Antonio Di Stasio, Shufang Zhu, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, Milda Zizyte

ContextQ: Generated Questions to Support Meaningful Parent-Child Dialogue While Co-Reading, *IDC 2024*
Griffin Dietz Smith, *Siddhartha Prasad*, Matt J Davidson, Leah Findlater, R Benjamin Shapiro

Forge: A Tool and Language for Teaching Formal Methods, *OOPSLA 2024*
Tim Nelson, Ben Greenman, *Siddhartha Prasad*, Tristan Dyer, Ethan Bove, Qianfan Chen, Charles Cutting, Thomas Del Vecchio, Sidney LeVine, Julianne Rudner, Ben Ryjikov, Alexander Varga, Andrew Wagner, Luke West, Shriram Krishnamurthi

Conceptual Mutation Testing for Student Programming Misconceptions, *‹Programming› 2024*
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Generating Programs Trivially: Student Use of Large Language Models, *CompEd 2023*
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Making Hay from Wheats: A Classsourcing Method to Identify Misconceptions, *Koli Calling 2022*
Siddhartha Prasad, Ben Greenman, Tim Nelson, John Wrenn, Shriram Krishnamurthi

Large-Scale Intelligent Microservices, *IEEE BigData 2020*
Mark Hamilton, Nick Gonsalves, Christina Lee, Anand Raman, Brendan Walsh, *Siddhartha Prasad*, Dalitso Banda, Lucy Zhang, Lei Zhang, William T Freeman

Teaching Assistant

Brown University

Logic for Systems — *2022–2026*

Programming Languages — *2025*

Tufts University

Programming Languages — *2015–2016*

Algorithms — *2014*

Service

2026 · Artifact Reviewer, The Art, Science, and Engineering of Programming

Affiliations & Certifications

Sheridan Teaching Seminar Certificate · Brown University
Tau Beta Pi, The Engineering Honor Society · Tufts University